

Proposed mechanistic hypothesis: ELF-EMF and Tumour Cell Death

Rationale and scope

The complete dataset contains 49 studies distributed in 439 experimental conditions, covering a wide spectrum of electromagnetic field modalities, each of them operates through distinct physical principles, targets different biological structures, and has been investigated under exposure conditions that vary by orders of magnitude in frequency, intensity, waveform, and duration. Even within the ELF-EMF category alone, which represents the largest macrocategory in the dataset, the range of tested frequencies spans from 0.01 Hz to 272 Hz, with intensities ranging from tens of nT to hundreds of mT. This makes it impossible to propose a single, coherent mechanistic hypothesis that would apply equally across all modalities. For this reason, the present proposal deliberately restricts its focus to one field category: ELF-EMF, as it is the category for which the evidence base is richest.

Restricting to the ELF-EMF doesn't imply that other modalities are less scientifically interesting, or that they do not deserve mechanistic investigation. ELF-EMF is where the available evidence is sufficient to support a mechanistic argument, and it is therefore where a first hypothesis can be formulated with some evidence.

It should also be known that not every mechanistic step proposed in this hypothesis is equally well supported. Where the evidence is direct and replicable, this is stated explicitly. Where a step is inferred from indirect observations, or where the precise biophysical trigger remains unknown, the uncertainty is acknowledged in the text. The goal here is to offer a structured and falsifiable framework that can guide experimental design and is honest about possible gaps.

Mechanistic hypothesis

Based on the experimental conditions reviewed in the literature and summarised in the data table below, the following mechanistic hypothesis is proposed. Three interconnected axes are identified, operating in parallel and converging on tumour cell death.

Axis 1 — Membrane perturbation, calcium dysregulation, and ROS accumulation

ELF-EMF exposure perturbs the plasma membrane and alters the activity of voltage-gated calcium channels (VGCC), leading to an increase in intracellular calcium concentration. This is supported by direct measurements across multiple cell lines. The altered intracellular calcium load subsequently perturbs mitochondrial membrane integrity, resulting in dysfunction of the mitochondrial electron transport chain and increased production of reactive oxygen species (ROS). Direct evidence for mitochondrial calcium perturbation is provided by evidence in the dataset. The resulting ROS accumulation overwhelms the cellular antioxidant defences (SOD, catalase) and reduces them after ELF-EMF exposure, as several studies in the corpus report. The oxidative stress triggers the intrinsic mitochondrial apoptotic pathway, leading to cytochrome c release, activation of caspases 3 and 7, and apoptosis. The specific link between oxidative stress and cytochrome c release in this context is an inferred step consistent with the established literature on the intrinsic apoptotic pathway.

In addition to apoptosis, the calcium and ROS dependent pathway may also engage alternative cell death modalities. It is hypothesised that necroptosis and ferroptosis may occur downstream of the same calcium dysregulation and oxidative stress described above, with their relative contribution depending on field parameters and cell type.

Axis 2 — Anti-Warburg metabolic shift (Warburg-phenotype cells only)

In tumor cells operating under the Warburg phenotype, characterised by suppressed mitochondrial oxidative phosphorylation (OXPHOS) and dependence on glycolysis, ELF-EMF exposure has been shown to reactivate mitochondrial respiration, inducing a metabolic shift toward OXPHOS. This anti-Warburg effect is directly supported by transcriptomic evidence from

the review studies, and it shows upregulation of COX2, COX4, MT-ATP6 and ATP5B. This effect was absent in non-tumorigenic cell lines, demonstrating intrinsic selectivity. The reactivation of previously suppressed mitochondria generates an additional ROS burden, which amplifies the oxidative stress described in Axis 1 and lowers the threshold for cell death in already-stressed tumour cells. This axis is not operative in cells with normal mitochondrial metabolism.

Axis 3 — p53 upregulation and mitochondrial apoptotic pathway (wild-type p53 only)

Independently of Axes 1 and 2, ELF-EMF exposure has been shown to upregulate p53 and modulate the expression of downstream targets including BAX, BCL-2, P21, and MDM2. In cells retaining wild-type p53 function, this upregulation shifts the BAX/BCL-2 balance toward apoptosis, activating the intrinsic mitochondrial pathway and resulting in cytochrome c release, caspase activation, and apoptosis. This axis is directly supported by genomic data in the table across multiple studies. In cells harbouring TP53 loss-of-function mutations, p53 upregulation does not translate into functional apoptotic signalling, and cell death, if it occurs, proceeds exclusively through Axes 1 and 2. Importantly, one study in the table noted that inactivation of p53 and pRb alone (via SV-40 Large T antigen) in a non-tumorigenic immortalised cell line was not sufficient to confer ELF-EMF susceptibility, suggesting that p53 status is not the primary determinant of tumor selectivity.

Evidentiary Basis

The table below summarises which steps of the hypothesis are directly supported by evidence in the reviewed studies, and which represent inferred links.

Mechanistic step	Status	Source
Plasma membrane perturbation → Ca^{2+} channel activation	Direct evidence	Dataset (Paper 16)
↑ Intracellular Ca^{2+} → mitochondrial Ca^{2+} perturbation	Direct evidence	Dataset (Paper 18)
Mitochondrial dysfunction → ↑ ROS	Direct evidence	Dataset (Papers 8, 9, 13)
↓ Antioxidant defences (SOD, catalase)	Direct evidence	Dataset (Paper 9)
Oxidative stress → cytochrome c release	Inferred	Consistent with established apoptosis literature
Cytochrome c → caspase 3/7 → apoptosis	Direct evidence	Dataset (Papers 3, 6)
Anti-Warburg shift → ↑ ROS (Warburg cells only)	Direct evidence	Dataset (Papers 2, 5)
p53 upregulation → BAX↑/BCL-2↓ → apoptosis	Direct evidence	Dataset (Papers 3, 6)
Tumour selectivity via Warburg phenotype	Direct evidence	Dataset (Paper 5, Paper 18)
Ca^{2+} + ROS → necroptosis (RIPK1/RIPK3/MLKL; abolished by verapamil and NAC)	Direct evidence	Dataset (Paper 25)

Ca ²⁺ + ROS → ferroptosis (partial reversal by ferrostatin-1)	Direct evidence	Dataset (Paper 9)
Necroptosis/ferroptosis as downstream of Axis 1 Ca ²⁺ /ROS pathway	Inferred	Mechanistic extension from Papers 25 and 9

Selectivity

Axis 1. Normal cells operate under conditions of comparatively low baseline oxidative stress, with antioxidant defence systems (superoxide dismutase, catalase, glutathione peroxidase) functioning very below their maximum capacity. On the other hand, tumour cells are characterised by chronically elevated ROS levels as a byproduct of their accelerated metabolism and mitochondrial dysfunction; their antioxidant systems are already partially engaged simply to maintain homeostasis. The consequence is that the same ELF-EMF-induced ROS increment pushes tumour cells beyond a critical threshold that normal cells can absorb without difficulty. The field does not create a different biochemical event in the two cell types; it creates the same event in cells that differ profoundly in how much margin they have left to handle it.

Axis 2. The anti-Warburg effect proposed here, that is, the reactivation of mitochondrial oxidative phosphorylation and the consequent additional ROS burden, is, by definition, inoperative in cells that already rely predominantly on OXPHOS for energy production. Normal differentiated cells have no suppressed mitochondrial respiration to reactivate. The field stimulus therefore finds no metabolic target in healthy tissue, and the amplification mechanism that makes Axis 2 potentially damaging in tumour cells simply does not engage.

Axis 3. In normal cells, p53 is functional and its upregulation primarily reinforces existing DNA repair and cell cycle arrest mechanisms rather than triggering apoptosis, because the level of accumulated damage does not reach the threshold required to commit the cell to programmed death. In tumour cells having wild type p53, the same upregulation occurs in the context of significantly greater oxidative and replicative stress, shifting the balance toward apoptotic signalling. In tumour cells harbouring loss-of-function TP53 mutations, the axis is altogether non-functional, and whatever selectivity exists must be attributed to Axes 1 and 2 alone.

However, it must be stated clearly that selectivity is not an inherent and unconditional property of ELF-EMF exposure. The dataset reviewed here shows that normal cells can be affected when exposure parameters fall outside an appropriate range, and that the differential response between tumour and healthy tissue is strongly dependent on the choice of the right combination of parameters. Selectivity is a target: identifying the parameter combinations that maximise the differential response between tumor and normal cells is precisely the experimental challenge that this hypothesis needs to face.

Parameter selection strategy

The candidate field conditions should be identified through computational modelling before any experiment is carried out. Given the demonstrated sensitivity of the biological response to small changes in frequency and intensity, and the shift in optimal frequency that occurs between 2D and 3D geometry for the same cell line, a blind experimental scan would be both inefficient and hard to interpret.

The proposed model operates in two coupled stages. The first simulates how an applied ELF-EMF field induces currents and transmembrane voltages as a function of frequency and intensity, using a three-dimensional spheroid geometry rather than a single flat cell, because the oxygen gradient, the spatial arrangement of mitochondria, and the layered electrical properties of a tumour mass all influence how the field is experienced at the membrane level. The second stage takes this induced perturbation as input and simulates the downstream response in terms of

cytoplasmic and mitochondrial calcium, mitochondrial membrane potential, and ROS accumulation over time.

The Warburg phenotype enters the model not as a dynamic variable but as a set of initial conditions: the tumour cell is parameterised with a lower resting mitochondrial membrane potential, higher basal ROS, and greater T-type calcium channel density, distinguishing it from the healthy cell model and producing a differential response to the same field input. The p53 axis lies entirely downstream of what the model tracks and remains a question for the laboratory, not for the simulation.

Such a model does not exist yet in the form described here and will need to be developed as part of this hypothesis; once built and validated, it will allow systematic sweeping of frequency and intensity across the ELF range, identifying prior to any experiment the parameter combinations most likely to drive tumour cells above a critical threshold while leaving normal cells unaffected.

Experimental design

1. Cell lines and culture. Breast cancer: MCF-7 (wild-type p53) and MDA-MB-231 (mutant p53), with MCF-10A as the non-tumorigenic control. Glioblastoma: U87 (wild-type p53) and U251 (mutant p53), with normal human astrocytes (NHA) as the healthy counterpart; U251-p53, a variant of U251 with p53 re-introduced, is also included to isolate the contribution of Axis 3. All lines are grown as 3D spheroids to better reproduce the real tumor mass.
2. Exposure setup. A Helmholtz coil system delivers a uniform field at the frequencies and intensities identified by the computational model, with a small static background field of approximately 45 μ T. The exposure chamber sits inside a mu-metal shield to eliminate stray environmental fields, and temperature is monitored continuously with a fibre-optic probe to exclude any thermal contribution. Every run includes both a sham condition and an untreated control.
3. Assays. Measurements are organised to follow the mechanistic chain proposed in this hypothesis.
 - a. Cell fate: viability (MTT); proliferation (BrdU and spheroid volume); apoptosis (Annexin V/PI); necrosis (PI, LDH release); necroptosis (RIPK1, RIPK3, MLKL phosphorylation, with necrostatin-1 as pathway control); ferroptosis (GPX4, lipid peroxidation via C11-BODIPY, with ferrostatin-1 as control); senescence (SA- β -gal, p21); motility (scratch and Transwell assays).
 - b. Mitochondria, redox and ions: total ROS (DCFDA) and mitochondrial superoxide (MitoSOX); antioxidant capacity (SOD1/2, catalase, GSH/GSSG); mitochondrial membrane potential $\Delta\psi_m$ (JC-1 or TMRE); cytoplasmic calcium (Fluo-4) and mitochondrial calcium (Rhod-2); cytochrome c release; metabolic flux (Seahorse: OCR and ECAR).
 - c. Morphology: spheroid confocal imaging for live/dead distribution and hypoxia mapping (HIF-1 α , pimonidazole); cytoskeletal organisation and membrane integrity assessed by fluorescence microscopy (phalloidin for actin, β -tubulin) and electron microscopy on a subset of samples.
 - d. Molecular targets (RT-qPCR and western blot, with RNA-sequence at the optimal condition):
 - i. for Axis 1: CACNA1G (Cav3.1), CACNA1H (Cav3.2), MCU;
 - ii. for Axis 2: COX2, COX4, MT-ATP6, ATP5B, HK2, LDHA, PDK1, GLUT1;
 - iii. for Axis 3: TP53, BAX, BAK, BCL2, cleaved caspases 3 and 7;
 - iv. for alternative death pathways: RIPK1, RIPK3, MLKL, GPX4, ACSL4.
4. Pharmacological controls. To confirm the mechanistic routes, selective inhibitors are applied in parallel arms: NAC to scavenge ROS, verapamil or mibefradil to block L- and T-type calcium channels, necrostatin-1 for necroptosis, ferrostatin-1 for ferroptosis, and

pifithrin- α to inhibit p53. Selectivity is quantified as the ratio of healthy-cell viability to cancer-cell viability across all tested conditions, and the parameter sets yielding the highest selectivity index are prioritised for further validation.

Predictions and falsification

If the hypothesis is correct, the following falsifiable predictions should hold.

P1 — The optimal frequency depends on geometry and is predicted by the model. The computational model should identify, for each cell line, a frequency window that produces a differential response between tumour and normal cells. If the experimental results fall outside those predicted windows, the parameter selection strategy has failed and the model requires revision.

P2 — Calcium and ROS are convergent and interdependent. At the optimal field condition, both cytoplasmic calcium and ROS should rise in tumour cells. Blocking calcium influx with verapamil or mibefradil should attenuate the ROS increase, and scavenging ROS with NAC should reduce cell death. If the two signals are independent, that is, if blocking one does not affect the other, Axis 1 as described here is wrong.

P3 — p53 status determines the mode of death, not the fact of it. Wild-type p53 lines (MCF-7, U87) should show predominantly apoptotic death, sensitive to pifithrin- α . Mutant p53 lines (MDA-MB-231, U251) should die through alternative pathways, sensitive to necrostatin-1 or ferrostatin-1 instead. Restoring p53 in U251-p53 should partially shift the death mode back toward apoptosis.

P4 — The anti-Warburg shift occurs only in tumour cells. Only cancer lines should show an increase in OCR and a decrease in ECAR following exposure, accompanied by upregulation of respiratory chain genes. Healthy cells should show no such metabolic reorganisation under the same conditions.

P5 — Selectivity is parameter-dependent, cell-line-specific, and has measurable limits. The selectivity index should be highest at the parameter combinations identified by the model for each cell line and should decrease outside those windows. Since the optimal frequency has been shown to vary between cell lines, selectivity is not expected to be universal, a condition that is selective for one tumour type may not be equally selective for another. The healthy cell lines included as controls should define, for each pairing, the boundary at which selectivity breaks down, providing a quantitative safety margin that is specific to the tumour type under investigation.